ASR Model Benchmark Report

FLEURS Bengali · 1,322 labeled utterances · concurrency 10 · all five models measured 2026-09-01

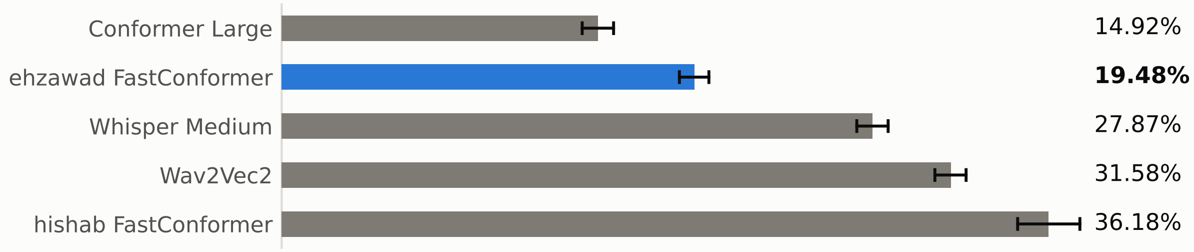
Supersedes the 2026-08-03 report: every figure below is a fresh measurement, not a carried-over number

The earlier report mixed three models benchmarked on an RTX 2050 and an RTX 3070. This one re-runs all five on one GPU, on the same rebuilt FLEURS set, through the same unmodified harness and the same audio path, each model loaded alone. Accuracy AND speed are therefore both controlled. Two of the earlier report's conclusions do not survive that.

CONFORMER LARGE BEST ACCURACY	EHZAWAD FASTCONFORMER ★ THIS REPO'S MODEL	WHISPER MEDIUM	WAV2VEC2	HISHAB FASTCONFORMER
14.9%	**19.5%**	**27.9%**	**31.6%**	**36.2%**
word error rate	word error rate	word error rate	word error rate	word error rate
CER 4.55%	CER 5.68%	CER 8.96%	CER 9.79%	CER 9.03%
Failed 0 / 1,322	Failed 0 / 1,322	Failed 0 / 1,322	Failed 0 / 1,322	Failed 0 / 1,322

Word Error Rate (WER)

corpus-level, lower is better · whiskers are the 95% bootstrap interval (10,000 utterance resamples)



Character Error Rate (CER)

same formula, at the character level

What is controlled

All five models ran on the same RTX 5080, through the same server process, the same audio path and the same unmodified harness, over the same 1,322 utterances, with the same failure policy (a failed request scores as an empty hypothesis, never dropped). Each was loaded alone, so none competed for VRAM with another. Accuracy and speed are both comparable across the whole table.

No two confidence intervals overlap, so the accuracy ordering is not sampling noise.

Cross-check on the newly added model

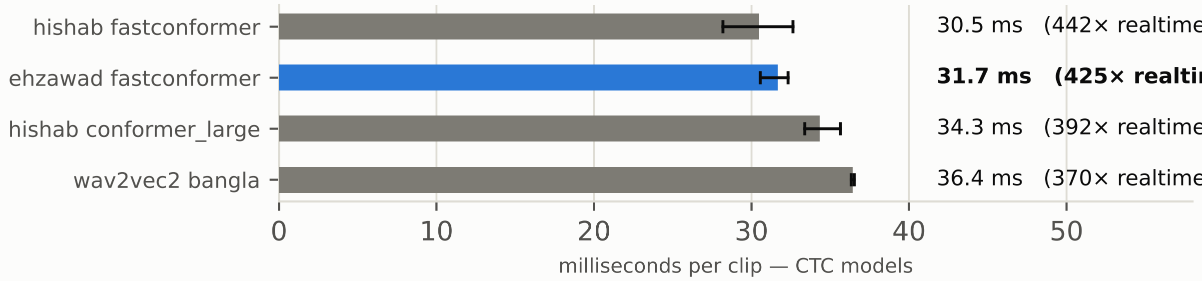
On the 920-utterance test split alone this run gives 19.65%, against the 19.57% published on the model card for that same split. Test and validation also score alike (19.65% / 19.08%); a model that had seen validation in training would score far better on it, which is evidence against contamination of the merged set.

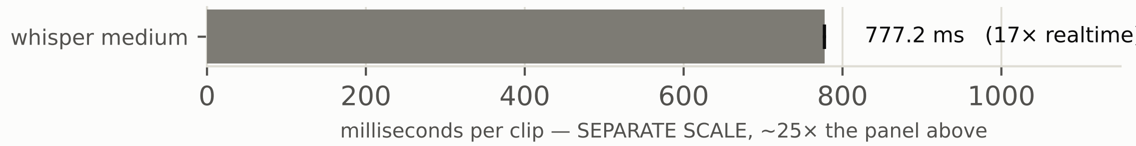
Speed

Two measurements. The isolated probe shows what the models themselves cost; the service benchmark shows what the deployed endpoint delivered, serving path included.

Isolated model compute — sequential, in-process, 3 repeats, one model resident at a time

60 identical clips, warm-up discarded. No HTTP, no base64, no queueing. Whiskers are min–max across repeats.





Autoregressive decoding costs about 25x, and it is not a serving artefact.

Whisper Medium spends 777 ms per clip against 30–36 ms for every CTC model here. It emits one token at a time through a decoder, where the CTC models emit a whole utterance in one forward pass. Among the CTC models the two FastConformers overlap and are tied; Conformer Large and Wav2Vec2 cost a little more.

Service throughput — the harness, at concurrency 10

Serialised endpoint: throughput is 1/service-time, and the latency below is mostly queueing behind the lock.

Off this scale: Whisper Medium 1.26 req/s (7.89s mean).

The four CTC models land within 14% of each other through the full serving path, compressing the compute gaps above: per-call overhead (HTTP, base64, file staging, a fixed transcribe() cost) is a large share of a ~30 ms request. Whisper is unaffected by that compression — at 777 ms of compute the overhead is noise, which is why it alone keeps its full disadvantage end-to-end.

MODEL	THROUGHPUT	MEAN	p50	p95	p99	PARAMS
Conformer Large	28.04 req/s	0.36s	0.34s	0.39s	0.50s	121.5M
ehzawad FastConformer	**29.39 req/s**	**0.34s**	**0.32s**	**0.37s**	**0.48s**	**115.6M**
Whisper Medium	1.26 req/s	7.89s	7.82s	9.53s	10.38s	763.9M
Wav2Vec2	26.26 req/s	0.38s	0.37s	0.43s	0.48s	315.5M
hishab FastConformer	29.89 req/s	0.33s	0.32s	0.36s	0.48s	115.6M

Wav2Vec2 and Whisper are the two largest models here and the two slowest.

At 315.5M and 763.9M parameters they are 2.7x and 6.6x the FastConformers, yet both are beaten on accuracy by the 121.5M Conformer Large — size is not buying accuracy here. (Reached over the network as the pre-existing Triton service, the same wav2vec2 weights return 13.53 req/s against 26.26 locally: transport, not the model.)

What changed against the 2026-08-03 report

Same models, same eval set, measured again under controlled conditions.

MODEL	WER THEN	WER NOW	Δ WER	req/s THEN → NOW	NOTE
Conformer Large	15.71%	14.92%	-0.79 pp	10.23 → 28.04	—
ehzawad FastConformer	**not benchmarked**	**19.48%**	**—**	**— → 29.39**	**newly added**
Whisper Medium	not benchmarked	27.87%	—	— → 1.26	newly added
Wav2Vec2	34.73%	31.58%	-3.15 pp	0.55 → 26.26	—
hishab FastConformer	37.36%	36.18%	-1.18 pp	20.84 → 29.89	—

Two conclusions from the earlier report do not survive the rerun.

THE SPEED TRADE-OFF LARGELY DISAPPEARS. That report measured Conformer Large at half FastConformer's throughput (10.23 vs 20.84 req/s) and recommended FastConformer for latency-sensitive paths on that basis. On one GPU the gap is 13% of model compute and 6% end-to-end — so the accuracy leader is no longer meaningfully the slow option.

WAV2VEC2'S OLD NUMBERS WERE ITS DEPLOYMENT, NOT ITS MODEL. The earlier report could only reach it as a remote service and recorded 34.73% WER at 0.55 req/s with 16 failures, disclaiming that this might reflect the deployment. It did. The service turned out to run SayedShaun/bangla-wave2vec2-unigram, exported to ONNX under Triton. Loading those same weights here in PyTorch gives 31.58% WER at 26.26 req/s with zero failures — 48x the throughput. Over the network today that service returns 31.52% at 13.53 req/s, so the residual gap is transport, not the model.

Model specs

MODEL	CHECKPOINT	PARAMS	GPU	FAILED
Conformer Large	hishab/titu_stt_bn_conformer_large	121.5M	RTX 5080	0 / 1,322
ehzawad FastConformer	**ehzawad/stt_bn_fastconformer**	**115.6M**	**RTX 5080**	**0 / 1,322**
Whisper Medium	SayedShaun/bengali-whisper-medium	763.9M	RTX 5080	0 / 1,322
Wav2Vec2	SayedShaun/bangla-wave2vec2-unigram	315.5M	RTX 5080	0 / 1,322
hishab FastConformer	hishab/titu_stt_bn_fastconformer	115.6M	RTX 5080	0 / 1,322

Ground truth: FLEURS (Google, CC-BY-4.0), bn_in config, test + validation merged — 1,322 utterances, rebuilt with the original run's own downloader. Wav2Vec2 and Whisper are the two backends of the pre-existing ASR service, identified from its conversion scripts and run here in PyTorch rather than ONNX or CTranslate2, so the comparison isolates the models rather than their runtimes; on a shared clip the local and remote wav2vec2 transcripts match exactly.

Detailed error breakdown

MODEL	W-SUB	W-INS	W-DEL	W-HITS	C-SUB	C-INS	C-DEL	C-HITS
Conformer Large	2,776	564	431	22,071	2,523	3,513	1,435	160,091
ehzawad FastConformer	**4,050**	**473**	**401**	**20,827**	**3,525**	**3,317**	**2,470**	**158,054**
Whisper Medium	6,177	587	281	18,820	6,560	3,420	4,712	152,777
Wav2Vec2	6,969	535	478	17,831	7,232	3,442	5,384	151,433
hishab FastConformer	5,330	3,695	120	19,828	5,204	7,230	2,375	156,470

The two FastConformers fail in opposite ways. The hishab checkpoint inserts 3,695 words against 120 deletions — an over-eager CTC decode that emits far more than it should. The ehzawad checkpoint is balanced at 473 insertions against 401 deletions, and Conformer Large likewise at 564 against 431. Insertion-heavy decoding is the single largest component of the worst score in this table. Wav2Vec2 fails differently again, skewing to substitutions — it hears words, but the wrong ones.

Method notes

WER/CER strip punctuation before alignment — FLEURS references carry quotes and the Bengali dari (danda) that no model transcribes, which would otherwise inflate errors with word-boundary artifacts rather than real mistakes.

All five runs used the harness unmodified and the same audio path (16 kHz mono, librosa/soxr resampling, pause-seeking segmentation above 25 s), so accuracy differences are attributable to the checkpoints. Intervals are 95% percentile bootstrap over 10,000 utterance resamples; one benchmark run per model, so throughput carries no interval of its own.

Takeaways

Conformer Large is the most accurate — and it is the smallest model but one.

14.92% WER (95% CI 14.18%-15.66%) against 19.48% for the ehzawad checkpoint, 27.87% for Whisper Medium, 31.58% for Wav2Vec2 and 36.18% for the hishab FastConformer. No two intervals overlap. At 121.5M parameters it beats a 763.9M Whisper by 13 points and a 315.5M Wav2Vec2 by 17 — on this benchmark, size is not buying accuracy.

It is also no longer the slow option — the old trade-off was hardware.

The 2026-08-03 report measured it at half FastConformer's throughput and recommended against it for latency-sensitive paths. Measured here it costs 13% in isolated model compute and 6% end-to-end. Paying 13% for 4.6 points of WER is a different decision from paying 2x, and it points the recommendation the other way.

The ehzawad checkpoint is second, and dominates the one it replaces.

19.48% against 36.18% — less than half the word error rate at the same architecture, the same parameter count and speed that overlaps within run-to-run variance. Against the hishab FastConformer it is a straight upgrade at no measured cost. Against Conformer Large it trades 4.6 points of WER for a little over 10% of compute.

Whisper Medium buys nothing here, and costs about 25x.

777 ms per clip against 30-36 ms for every CTC model, because it decodes one token at a time rather than emitting an utterance in a single forward pass. That is architectural, not a serving artefact — it is the one model whose disadvantage survives end-to-end (1.26 req/s against 26-30). It also places third on accuracy, so Conformer Large beats it on both axes at a sixth of the size.

What this still does not establish.

FLEURS is short read speech. It says nothing about conversational, noisy, dialectal or long-form audio — and the ehzawad card itself reports roughly 5 points worse WER on spontaneous speech. Whisper in particular is built for long-form and multi-domain audio, so a benchmark of short read clips is close to its least favourable case. Nor does a serialised endpoint predict behaviour under a batching server at production concurrency.

Recommendation

Default to Conformer Large (hishab/titu_stt_bn_conformer_large). Most accurate by 4.6 points, at 13% of model compute on this hardware — a trade the earlier report's numbers argued against and these do not.
Keep the ehzawad checkpoint for genuinely latency-bound paths: it is the accuracy runner-up at FastConformer speed.
Retire the hishab FastConformer, Wav2Vec2 and Whisper Medium for this workload: the first is dominated outright by the ehzawad checkpoint, and the other two are larger, slower and less accurate than a 121.5M CTC model. Whisper may still earn its place on long-form or multi-domain audio, which this benchmark does not test.
Then re-measure the two leaders under your real serving stack and on in-domain audio before committing — the measurement this report deliberately does not claim to make.

Reproducing this

```
<pipeline>/scripts/download_eval_data.py --data-dir eval_fleurs_bn    # the original downloader
./run_all_benchmarks.sh     # all five models, one GPU, one resident at a time
python speed_probe.py       # isolated per-model compute
python collect_results.py && python make_report.py     # summary.json, then this PDF
```